

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The positive number a is **230%** of the number b , and a is **60%** of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

Answer

- A. 138
- B. 217
- C. 283
- D. 383

Correct Answer: D

Rationale

Choice D is correct. It's given that a is **230%** of b . It follows that $a = \frac{230}{100}b$. It's also given that a is **60%** of c . It follows that $a = \frac{60}{100}c$. Since $a = \frac{230}{100}b$ and $a = \frac{60}{100}c$, it follows that $\frac{230}{100}b = \frac{60}{100}c$. Multiplying each side of this equation by $\frac{100}{60}$ yields $\frac{23}{6}b = c$. If c is $p\%$ of b , then $c = \frac{p}{100}b$. It follows that $\frac{23}{6} = \frac{p}{100}$. Multiplying each side of this equation by **100** yields $\frac{2,300}{6} = p$. It follows that the value of p is approximately **383.33**. Therefore, of the given choices, **383** is closest to the value of p .

Choice A is incorrect. This is closest to the value of p if b is **230%** of a , rather than if a is **230%** of b , and if b is $p\%$ of c , rather than if c is $p\%$ of b .

Choice B is incorrect. This is closest to the value of p if a is **230%** greater than b , rather than **230%** of b .

Choice C is incorrect. This is closest to the value of p if c is $p\%$ greater than b , rather than $p\%$ of b .